PMx Weekly Digest: May 26, 2026

Content Source: Full Text

Paper 1: Model-Informed Dosing Regimen of Sugemalimab for European Patients With Non-Smal

Bibliographic Information

- **Title**: Model-Informed Dosing Regimen of Sugemalimab for European Patients With Non-Small Cell Lung Cancer: Bridging From Asian Clinical Data
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- **Journal**: Clinical Pharmacology & Therapeutics, 2025
- **DOI**: N/A

Paper Type: POPK_PKPD

Quick Take

This paper demonstrates a model-informed drug development (MIDD) strategy to bridge sugemalimab dosing from Asian to European NSCLC patients. Using a validated popPK model and exposure-response simulations, the authors identified that patients > 115 kg require a dose increase from 1200 mg to 1500 mg Q3W to match pivotal trial exposures. The analysis was accepted by EMA and MHRA for regulatory approval without additional trials—a landmark example of using pharmacometrics for global dose bridging.

Executive Summary

This study employed a rigorous MIDD framework to determine whether the fixed-dose regimen of sugemalimab 1200 mg Q3W, approved in China based on an all-Asian pivotal trial, provides adequate exposure in European NSCLC patients who generally have higher body weights. A popPK model built from 1002 subjects (97.6% Asian) across six trials was externally validated using data from a higher-dose regimen (1800 mg Q4W). PK simulations for patients weighing 80– 150 kg under three dosing scenarios (1200, 1500, 1800 mg Q3W) were compared to exposures from the GEMSTONE-302 study. The analysis demonstrated that 1200 mg Q3W maintained adequate exposure up to 115 kg, but patients 115– 150 kg required 1500 mg Q3W to achieve comparable C_{trough}, AUC, and C_{max}. Exposure-response models for PFS and OS confirmed that the proposed weight-based adjustment produces survival probabilities overlapping with those from Asian patients. Notably, the MIDD approach directly led to regulatory approvals by the EMA and MHRA without additional clinical trials, highlighting growing regulatory confidence in pharmacometric bridging strategies for ethnically insensitive biologics.

Scientific Context & Motivation

Monoclonal antibodies like sugemalimab are known to have linear PK, low ethnic sensitivity, and elimination primarily via non-specific proteolytic catabolism, which supports the ICH E5/E17 framework for using foreign clinical data. However, body weight significantly impacts monoclonal antibody clearance, and European patients have substantially higher average body weights ($\sim$ 70 kg for females, $\sim$ 80 kg for males) compared to the Asian pivotal study population (mean $\sim$ 62 kg, range 41– 96 kg). The key knowledge gap addressed was whether the approved 1200 mg Q3W fixed dose, which was studied almost exclusively in Asian patients, provides clinically adequate exposure in heavier European patients, and if not, what alternative dose and body weight threshold would restore comparable exposure and efficacy. This work challenges the one-size-fits-all assumption of fixed dosing for PD-L1 inhibitors and provides a quantitative framework for ethnic bridging that goes beyond traditional population PK comparisons by incorporating exposure-response modeling for efficacy endpoints.

⚡ Methodological Snapshot

The popPK model for sugemalimab was developed from 1002 subjects (97.6% Asian) across six clinical trials. The model structure is a two-compartment model with time-varying clearance, including body weight as a covariate on CL_0 and V_c. External validation was performed using data from 13 Asian patients receiving 1800 mg Q4W, evaluated via prediction error metrics and VPC. For dose bridging, individual EBEs from the model were used to simulate PK profiles for virtual patients with body weights uniformly sampled from 80– 150 kg (14 strata). Three dose levels (1200, 1500, 1800 mg Q3W) were compared to reference exposures from GEMSTONE-302 using $90\% CI$ of GMR with 80– 125% acceptance bounds. Exposure-response models for PFS (modified Gompertz) and OS (exponential hazard) were developed from GEMSTONE-302 data, using C_{trough} from Cycle 1 as the exposure metric. Survival simulations incorporated both between-subject variability in exposure and parameter uncertainty.

Structural Model Breakdown

The popPK model is a two-compartment model with time-varying clearance. The structural model includes: central compartment (V_c), peripheral compartment (V_p), intercompartmental clearance (Q), baseline clearance (CL_0), and clearance decay rate (K_{des}). Time-varying CL is modeled as $CL(t) = CL_0 \cdot \exp(-K_{des} \cdot t)$. Body weight influences CL_0 and V_c through allometric power relationships. The model includes inter-individual variability (IIV) on $CL_0, V_c, V_p, Q,$ and K_{des}, and inter-occasion variability (IOV) on CL. Residual variability is modeled as proportional plus additive error.

Detailed Methodological Analysis

Modeling Approach

Two-compartment population PK model with time-varying clearance (exponential decay), estimated using NONMEM. Covariates included body weight on CL_0 and V_c. Exposure-response models: modified Gompertz for PFS, exponential hazard for OS. All software: NONMEM 7.5.0, PsN 5.3.1, R 4.4 with mrgsolve.

Data Sources

Six clinical trials (2 Phase I, 2 Phase II, 2 Phase III) totaling 1002 subjects (97.6% Asian, 97.5% receiving 1200 mg Q3W). External validation: 13 Asian CRC patients receiving 1800 mg Q4W (146 PK samples). GEMSTONE-302: 320 NSCLC patients for ER modeling. Exposure-safety: 842 patients from 4 studies.

Estimation Methods

NONMEM with FOCE (First-Order Conditional Estimation) method for popPK model. Post hoc EBEs for individual parameters. Sensitivity analysis using η-sampling from conditional distributions.

Model Evaluation

External validation using prediction error metrics ($MPE, MAPE, F_{20}, F_{30}$), goodness-of-fit plots, and VPC. Reference-corrected VPC for extrapolation to overweight patients. ER models evaluated using VPC (Figure S3).

Covariate Analysis

Body weight was the primary covariate of interest, included on CL_0 and V_c with power exponents estimated from data. Other covariates (not specified in detail) were evaluated but found not clinically meaningful in the original model development. For simulations, only body weight was varied; other covariates were sampled with replacement from GEMSTONE-302 subjects.

Statistical Rigor Assessment

The analysis demonstrates appropriate statistical methods overall. The external validation metrics ($MPE, MAPE, F_{20}, F_{30}$) follow established acceptance criteria from the literature. The GMR-based approach with $90\% CIs$ is statistically rigorous and aligns with bioequivalence principles. Sensitivity analyses addressing η-shrinkage by comparing EBE-based and η-sampling methods show robustness. The exposure-response models incorporate parameter uncertainty via sampling from the asymptotic variance-covariance matrix. Limitations include the small external validation sample ($n = 13$), the single-dose-level ER model, and potential OS bias from crossover. The uniform weight distribution for simulation is not the true European distribution but was justified for exploring the full weight range. Missing data handling is not explicitly discussed but is less relevant as the simulation-based analysis uses complete simulated datasets.

Key Findings

The popPK model was successfully externally validated using 146 PK samples from 13 patients receiving 1800 mg Q4W, with acceptable prediction error metrics ($MPE - 7.84\%$ for $PRED$, $MAPE 17.14\%$, $F_{20} 80.14\%$). PK simulations revealed that the 1200 mg Q3W dose maintains adequate exposure ($90\% CI$ of $GMR > 0.80$ for $C_{trough}, C_{max},$ and AUC) for patients up to 115 kg, with $> 89\%$ of patients exceeding the clinical reference C_{trough} of 42.6 $\mu g/mL$. For patients weighing 115– 150 kg, $C_{trough} GMR$ fell to 0.81– 0.70, indicating significant underexposure. The 1500 mg Q3W dose for this subgroup restored $C_{trough} GMR$ to 1.01– 0.88, with all

GMR 90%CIs within 80%– 125%. The 1800 mg dose produced excessive exposure ($GMR >$ 1.25) for most weight strata below 135 kg. Efficacy simulations showed that the proposed regimen ($\leq$ 115 kg: 1200 mg; $>$ 115 kg: 1500 mg Q3W) produced survival probabilities similar to Asian patients, with median differences at 12 months of only 4.2% (PFS-investigator), 5.8% (PFS-BICR), and 8.6% (OS). Importantly, no exposure-safety relationship was identified across a wide dose range (3– 40 mg/kg Q3W).

Clinical & Regulatory Implications

The study provides clear dosing recommendations: European NSCLC patients weighing $\leq$ 115 kg should receive sugemalimab 1200 mg Q3W, while those $>$ 115 kg should receive 1500 mg Q3W. This weight-based threshold is practical for clinical implementation and is expected to maintain comparable efficacy to that observed in the pivotal Asian trial. Importantly, safety is projected to be similar as the 25% increase in C_{max} is within accepted bounds for PD-(L)1 class antibodies and no exposure-safety relationship was identified. The MIDD framework directly supported regulatory approvals by EMA (July 2024) and MHRA (October 2024) without requiring additional clinical trials, demonstrating significant cost and time savings. For special populations, the analysis suggests that weight (not ethnicity per se) is the key determinant of appropriate dosing, consistent with ICH E5/E17 principles.

Strengths & Limitations

Strengths

- Comprehensive MIDD framework integrating popPK, external validation, PK simulation, and exposure-response modeling for both efficacy and safety
- External validation of the popPK model using a different dosing regimen (1800 mg Q4W), demonstrating robustness for extrapolation beyond the model-building dose range
- Rigorous dose-adjustment criteria adapted from FDA PD-(L)1 alternative dosing guidance, including both central tendency and variability via 90%CI of GMR
- Sensitivity analyses addressing η-shrinkage by comparing EBE-based and η-sampling simulation approaches, showing consistent results
- Use of Cycle 1 exposure metrics to avoid response-driven confounding in exposure-response analyses
- Successful translation to regulatory approval (EMA and MHRA), providing real-world validation of the approach

Limitations (Acknowledged by Authors)

- Only 75 subjects (7.5%) in the model-building dataset weighed $\geq$ 80 kg, limiting direct inference for heavy patients
- High η-shrinkage for $K_{des}, CL_T,$ and V_p (39.5%– 49.6%) may attenuate variability in simulated exposures
- ER model was developed from a single dose level (1200 mg Q3W), limiting ability to reliably quantify the exposure-efficacy relationship for extrapolation

- OS model may be biased due to crossover of 45 placebo patients receiving sugemalimab after progression
- No relationship identified between exposure and safety, which precluded formal safety modeling for the proposed dose
- Only Cycle 1 exposures were evaluated; steady-state C_{max} was not assessed per FDA guidance

Limitations (Expert Review)

- The external validation dataset comprised only 13 Asian patients; a more ethnically diverse external dataset would strengthen the extrapolation claim
- The reference C_{trough} threshold (42.6 $\mu g/mL$, the 5^{th} percentile of GEMSTONE-302) may be a conservative anchor; using a target exposure associated with a specific efficacy threshold would be more informative
- The simulation of body weight using uniform distributions within strata does not account for the actual European body weight distribution; real-world weight distributions would affect the proportion of patients needing dose adjustment
- No formal model-informed precision dosing framework was proposed; the binary threshold ($>$ 115 kg) may oversimplify continuous weight-exposure relationships
- The impact of immunogenicity (ADA) was acknowledged but not modeled; ADA could increase clearance and further reduce exposure in heavier patients, but this was considered conservative
- The bridging strategy assumes identical PK and PD between Asian and European populations beyond weight effects, which may not fully capture unmeasured ethnic differences in Fc receptor biology or target expression

Generalizability

The findings are specific to sugemalimab in NSCLC and may not directly extend to other PD-L1 inhibitors or indications. However, the MIDD framework—including external validation, PK simulation across weight strata, and exposure-response confirmation—is highly generalizable. The authors appropriately note that many PD-(L)1 antibodies show minimal ethnic sensitivity and similar weight-exposure relationships, supporting the broader applicability of this weight-based bridging approach for biologics with linear PK and wide therapeutic windows. The proposed 115 kg threshold and 1500 mg dose are specific to sugemalimab's PK properties and exposure-response characteristics.

Key Equations

Time-varying clearance model

$$CL(t) = CL_0 \cdot e^{-K_{des} \cdot t}$$

Sugemalimab clearance decreases over time, modeled with an exponential decay function where CL0 is the baseline clearance and Kdes is the rate constant for clearance decay.

Body weight effect on clearance

$$CL_0 = CL_{0,\text{ pop}} \cdot \left(\frac{WT}{WT_{\text{median}}}\right)^{\theta_{\text{CL}}} \cdot e^{\eta_{CL}}$$

Allometric relationship between body weight and baseline clearance, where CL0,pop is the typical CL0 at median weight, and θCL is the power exponent estimated from data.

Body weight effect on central volume

$$V_c = V_{c,\text{ pop}} \cdot \left(\frac{WT}{WT_{\text{median}}}\right)^{\theta_{Vc}} \cdot e^{\eta_{Vc}}$$

Allometric relationship between body weight and central volume of distribution, similar in structure to the clearance model.

Exponential hazard model for OS

$$h(t) = h_0 \cdot e^{\beta \cdot C_{\text{trough},C1}}$$

The hazard of death decreases exponentially with increasing Cycle 1 trough concentration, where h0 is the baseline hazard and β is the slope of exposure effect.

Modified Gompertz model for PFS

$$S(t) = e^{-\lambda \cdot (1-e^{-\alpha \cdot t}) \cdot e^{-\beta \cdot C_{\text{trough},C1}}}$$

Survival function for PFS incorporating exposure effect, where λ and α are shape parameters and β is the exposure-response coefficient.

Figures & Tables

- **Figure 1**: External validation of the popPK model: (a) Goodness-of-fit plots showing observed vs. predicted concentrations, and (b) Visual predictive check (VPC) for the 1800 mg Q4W regimen.
  - *Significance*: Demonstrates the model's ability to accurately predict concentrations for a dosing regimen (1800 mg Q4W) not used in model development, establishing credibility for extrapolation to higher doses.
- **Figure 2**: Boxplots of simulated Cycle 1 exposure metrics (C_{trough}, AUC, C_{max}) for sugemalimab 1200, 1500, and 1800 mg Q3W across 14 weight strata (80– 150 kg).
  - *Significance*: Central figure supporting dose adjustment recommendation; visually demonstrates the monotonic decrease in exposure with increasing body weight and the ability of higher doses to restore exposure.
- **Figure 3**: Survival probability curves from exposure-response simulations for PFS (investigator-assessed and BICR) and OS across three groups: Asian patients (1200 mg), 80– 115 kg patients (1200 mg), and 115– 150 kg patients (1500 mg).
  - *Significance*: Provides clinical context for the proposed dose adjustment by showing that survival outcomes for the high-weight/high-dose group overlap with those of Asian patients, supporting non-inferior efficacy.

- **Table 1**: External validation metrics: $MPE, MAPE, F_{20},$ and F_{30} for both $PRED$ and $IPRED$-based predictions of the 1800 mg Q4W dataset.
  - *Significance*: Quantifies model performance in extrapolation; all metrics fall within acceptable ranges, confirming the model's suitability for dose simulation.
- **Table 2**: Detailed simulation results across 42 scenarios (14 weight strata × 3 dose levels) showing percentage above clinical reference and GMR (90%CI) for $C_{trough}, AUC,$ and C_{max}.
  - *Significance*: Comprehensive data table used to identify the 115 kg threshold and justify the 1500 mg dose; shaded cells indicate GMR within 80–125% bounds.
- **Table 3**: Predicted PFS and OS survival probabilities at 12 and 24 months with $5^{th}, 50^{th},$ and 95^{th} percentiles for the three virtual patient groups.
  - *Significance*: Quantifies the predicted clinical impact of the proposed dosing strategy and demonstrates overlap in confidence intervals across groups.

Code & Reproducibility Assessment

The authors state that NONMEM 7.5.0, PsN 5.3.1, and R 4.4 (mrgsolve package) were used. No publicly available code repository or data sharing is mentioned. The popPK model and ER model equations are referenced but not fully detailed in the manuscript, limiting full reproducibility.

Future Directions

Several extensions are warranted: (1) Prospective evaluation of the proposed 1500 mg Q3W regimen in a real-world European cohort to confirm PK exposure and clinical outcomes; (2) Modeling of the impact of ADA on long-term exposure in heavier patients, which was conservatively omitted; (3) Application of the MIDD framework to other PD-(L)1 antibodies requiring similar weight-based bridging; (4) Exploration of weight-based dosing (mg/kg) versus fixed-dose with a threshold, including cost-benefit analyses; (5) Development of a model-informed precision dosing tool for sugemalimab that accounts for body weight, albumin, and other covariates; (6) Formal exposure-safety modeling using a larger dataset or meta-analysis of PD-(L)1 antibodies to establish safety bounds for exposure increases above $+25\%$.

Expert Commentary

This paper represents a textbook example of how pharmacometrics can directly support global drug approval and patient access. As a senior pharmacometrician, I find several aspects particularly commendable. First, the external validation using a different dose regimen (1800 mg Q4W) is a crucial step often omitted in similar bridging analyses—it provides genuine confidence in the model's extrapolative capacity. Second, the use of 90%CI of GMR with 80–125% bounds, adapted from bioequivalence principles, is a rigorous and regulator-friendly approach that explicitly addresses both central tendency and variability. Third, the sensitivity analysis on η -shrinkage demonstrates methodological maturity—many analyses would have stopped at EBE-based simulations without acknowledging the potential attenuation of variability. The choice of Cycle 1 exposure to avoid response-driven confounding is essential for monoclonal antibodies with time-varying clearance, and the authors correctly justify this decision against FDA's recommendation for steady-state C_{max}. One area where I would have liked to see more discussion is the clinical implementation: will the 115 kg threshold be practical in real-world clinics, and how will it be monitored? Also, while the uniform weight distribution approach is reasonable for exploring the full range, incorporating the actual European weight distribution would provide more realistic estimates of how many patients actually need dose adjustment. The conclusion that 1500 mg for > 115 kg 'restores' exposure to Asian levels is well-supported by the simulation data, and the modest differences in predicted survival ($< 10\%$) are reassuring for clinicians and regulators. This paper should be a required reading in pharmacometrics training programs for its clear demonstration of the MIDD paradigm.

Bottom Line

For practicing pharmacometricians, this paper exemplifies a rigorous, regulatory-accepted MIDD workflow for ethnic bridging: external validation of an established popPK model, systematic weight-stratified PK simulations with well-defined bioequivalence-inspired acceptance criteria, and exposure-response confirmation of clinical outcomes. The 115 kg threshold for dose escalation to 1500 mg Q3W is scientifically justified and has already received regulatory approval. Key methodological takeaways include the importance of using Cycle 1 exposure to avoid response-driven bias in ER modeling, addressing η-shrinkage through sensitivity analyses, and employing both central tendency and distributional criteria ($90\% CI$ of GMR) rather than simple mean comparisons. This work should be a reference case for future submissions requiring dose bridging across populations without dedicated clinical trials.

Figures

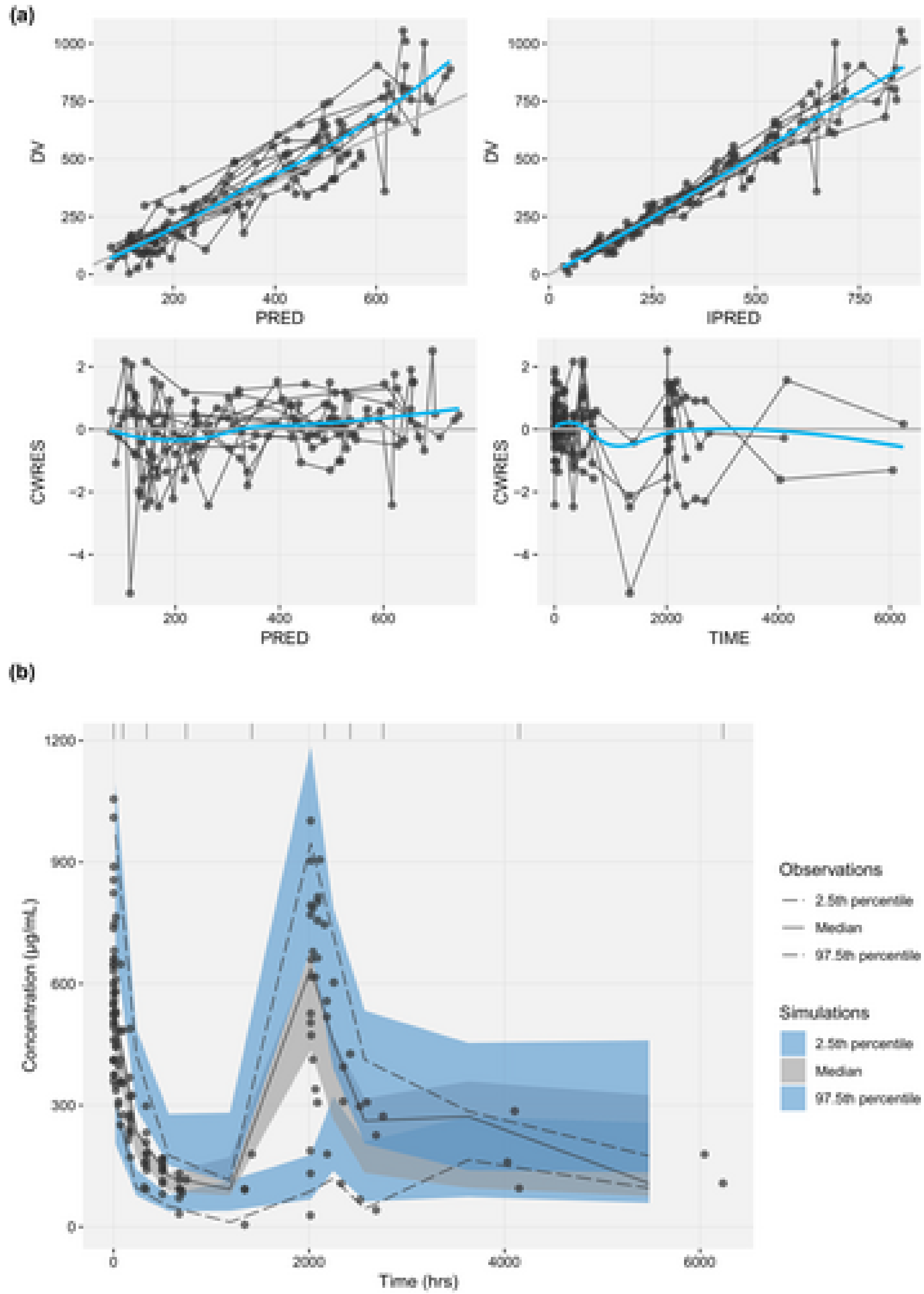

Figure 1: External validation for the population pharmacokinetic model. (a) Goodness-of-fit plots. (b) Visual predictive check.

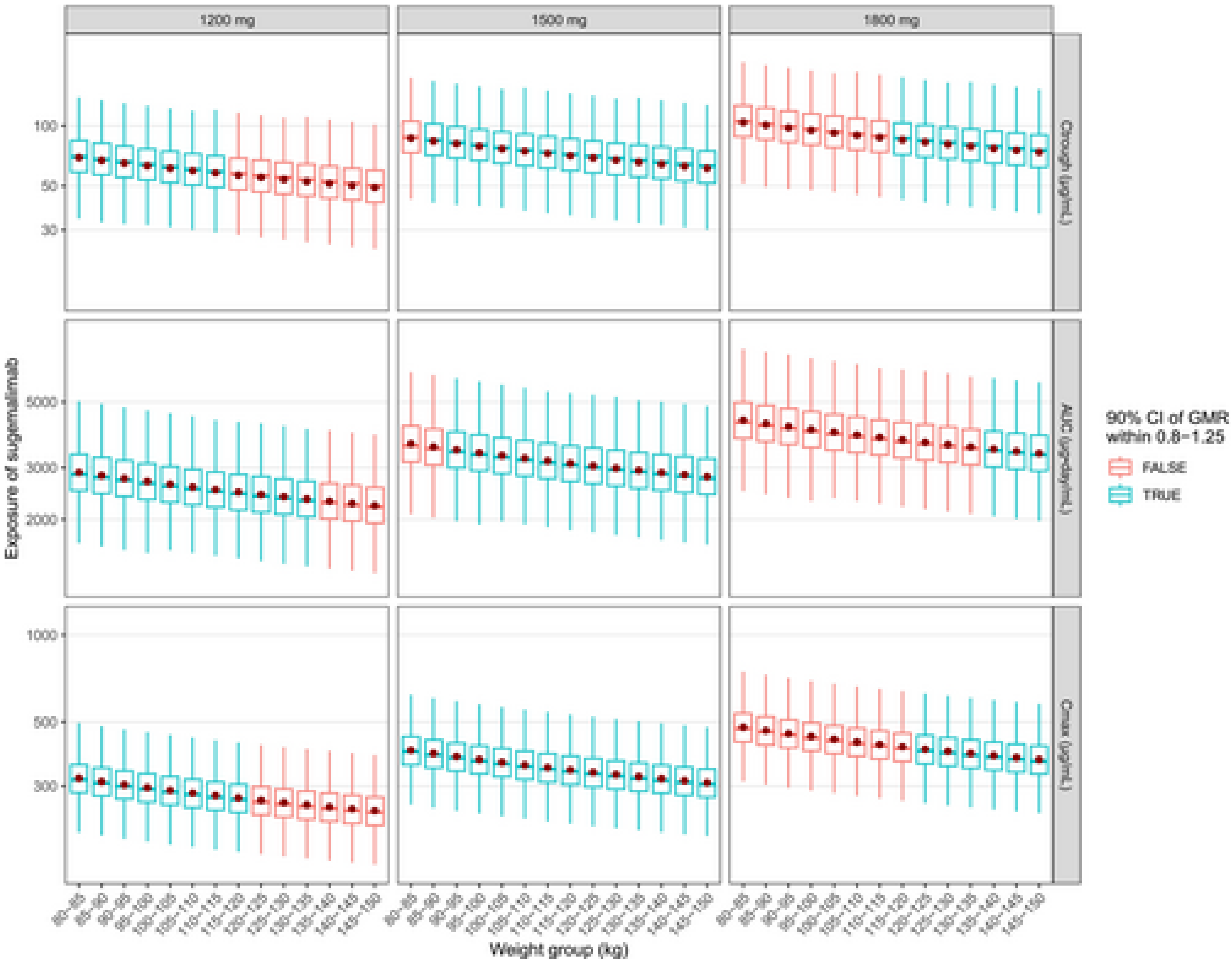

Figure 2: Boxplots of simulated exposure (Ctrough, AUC, and Cmaxof Cycle 1) for sugemalimab 1200, 1500, and 1800 mg Q3W dosing regimens from high body weight strata. Red i

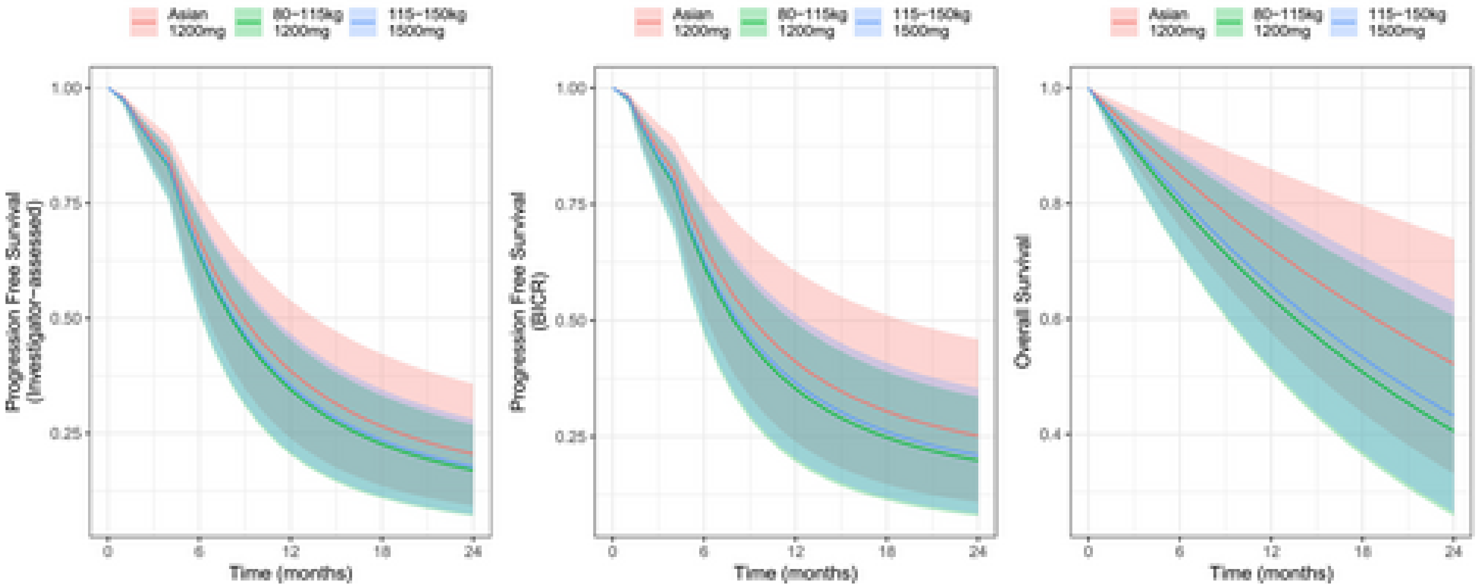

Figure 3: Survival probability of virtual patients from exposure-response for progression free survival (investigator-assessed), progression free survival (BICR), and over